BIDENGINE

AI Tender Intelligence & Bid Win Prediction Engine

Complete System Report

RFP/RFQ parsing, compliance evidence workflow, draft proposal generation, and explainable win probability scoring

Core Output	What the system produces
Opportunity Overview	Client, title, sector, deadline, budget, currency, selection method
Compliance Matrix	Mandatory requirements, evidence needed, pass/gap status
Draft Responses	AI-generated proposal sections using Claude with no-fabrication rules
Win Probability	Evidence-aware Bid / No Bid / Needs Evidence Review recommendation

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1. Executive Summary

BidEngine is an AI tender intelligence platform designed to help bidders understand RFP/RFQ documents, complete mandatory compliance evidence, draft proposal responses, and estimate win probability before submission. The system converts an uploaded procurement document into a structured bid workspace containing opportunity metadata, compliance checklist, evidence status, draft response questions, and an explainable Bid / No Bid recommendation.

The current implementation is a React/Vite web application with a client-side document parser, deterministic rule-based extraction, a lightweight historical bid scoring engine, and Claude-based proposal drafting. The model does not blindly predict a win. It first checks whether mandatory evidence is complete; if evidence is missing, the output stays in Pending Evidence Review until the bidder marks evidence as provided or identifies gaps.

Key idea
The engine treats compliance as a gate before win probability. This prevents misleading predictions when the bidder has not yet proven eligibility or mandatory capability.

2. Problem Understanding

In real procurement, bidders often lose time reading long RFPs, missing mandatory documents, or preparing generic responses that do not match evaluation criteria. A bid team needs a fast way to answer three questions:

- What is this opportunity about?
- What must we prove to qualify?
- Based on our evidence and past bid history, should we bid or not?

The project therefore focuses on turning an unstructured RFP/RFQ PDF into a decision-support workspace. The system is not only a summarizer; it is a bid-readiness and proposal-generation engine.

3. What Was Built

The system built is a full bid-response workflow application. It supports PDF, DOCX, and TXT uploads, extracts the tender content, classifies the opportunity, builds a compliance matrix, asks the bidder to complete evidence, generates draft proposal responses, and exports a bid response pack as PDF/Word-style report.

Module	What it does
Upload and Parsing	Reads PDF/DOCX/TXT tender files and converts them into clean text for analysis.
Metadata Extraction	Detects RFP/RFQ number, title, client, sector, deadline, budget, currency, pages, and selection method.
Requirement Extraction	Finds mandatory eligibility, technical, financial, SLA, training, reporting, and declaration requirements.
Compliance Workspace	Shows each requirement with evidence needed and lets the bidder mark Pass, Gap, or reset.
Win Probability Engine	Uses compliance, gaps, proposal quality, sector win rate, k-NN similar bids, and logistic-style scoring.
Draft Response Engine	Creates tailored proposal answers using Claude API or offline fallback templates.
Export Pack	Generates a structured bid response report including overview, checklist, win analysis, similar bids, and drafts.

4. Technology Stack and Components

Layer	Technology / Method	Reason for Use
Frontend	React + Vite	Fast demo development, interactive workspaces, local file upload, and easy deployment.
PDF Parsing	pdf.js	Runs in browser and extracts selectable PDF text without a backend.
DOCX Parsing	Mammoth.js	Extracts clean raw text from Word documents.
Historical Data	Embedded 120 bid history records	Used for hackathon demo scoring and similar bid comparison.
LLM Drafting	Claude Sonnet 4.5 via Anthropic API	Generates strong, professional proposal drafts while following no-fabrication rules.
Scoring Model	Hybrid heuristic + k-NN + logistic-style baseline	Lightweight, explainable, and suitable for 120-row dataset and client-side execution.
Export	Browser-generated PDF/Word-style pack	Allows judges/users to download final bid response packs.

Security note

For demo, Claude API is called from the browser using VITE_CLAUDE_API_KEY. In production, this should move to a backend API so the key is never exposed to users.

5. End-to-End Engine Workflow

Step	Engine Action	Output
1	User uploads an RFP/RFQ/Tender PDF,	Clean document text and page count.

	DOCX, or TXT.	
2	Parser extracts opportunity metadata.	Client, title, sector, deadline, budget, currency, selection method.
3	Requirement extractor identifies mandatory and scored requirements.	Compliance checklist with evidence needed.
4	Draft question generator converts scope/evaluation criteria into proposal questions.	Q1, Q2, Q3 style draft response sections.
5	Bidder marks evidence as provided or gap.	Compliance percentage and gaps found.
6	Win probability engine calculates bid score.	Probability, risk, recommendation, similar bids, improvement suggestions.
7	Claude generates tailored draft responses.	Editable proposal answers with placeholders where evidence is missing.
8	User exports final pack.	PDF/Word-style bid response pack.

6. RFP/RFQ Parsing and Extraction Logic

The parser is built as a layered engine instead of a single prompt. This is important because RFPs differ widely in layout. Some contain explicit questionnaire sections, while others contain tables, TOR, scope of work, SLA terms, annexures, or evaluation criteria. The system therefore combines deterministic extraction rules with optional LLM parsing.

Extraction Area	How It Works
Document number	Regex patterns detect RFQ Number, RFP Number, RFQ-006, RFP No. 166, and similar formats.
Title	Prioritizes Title of RFP/RFQ and cover-page titles; avoids pulling About/Background paragraphs into the title.
Deadline	Searches Closing date, Submission Deadline, Due Date, and no later than phrases.
Sector	Classifies by scope keywords, e.g., routers/Cisco/network -> Network Infrastructure, radio/campaign -> Media/Communications, portal/cloud/cybersecurity -> IT Services.
Requirements	Uses domain-specific templates for routers, radio airtime, awareness campaigns, and IT services, then falls back to generic requirement extraction.
Questions	Uses existing questionnaire sections where present; otherwise converts scope/evaluation criteria into proposal-writing questions.

This approach was chosen because pure LLM extraction can hallucinate, skip tables, or convert administrative text into requirements. Deterministic parsing keeps mandatory compliance stable, while Claude is used later for writing quality rather than for final eligibility decisions.

7. Compliance Evidence Workflow

Compliance is the core gate of the system. The engine extracts mandatory requirements but does not assume the bidder satisfies them. Each requirement starts as Needs Evidence unless the bidder marks evidence as provided or identifies it as a gap.

Status	Meaning	Effect on Win Probability
Needs Evidence	Requirement extracted but bidder has not yet provided or confirmed evidence.	Final probability is paused as Pending Evidence.
Pass	Bidder has evidence such as certificate, registration, portfolio, CV, BOQ, SLA, or annexure.	Increases compliance score and lowers gaps.
Gap	Mandatory evidence is missing or requirement cannot be met.	Creates mandatory penalty and may result in No Bid / Fix Gaps.
Reset	Clears the status back to Needs Evidence.	Returns the item to evidence review.

Examples of evidence include company registration, FBR/ATL proof, tax returns, OEM letters, Cisco certificates, portfolios, client references, team CVs, SLA commitments, implementation plans, financial templates, and signed annexures. The evidence workflow makes the model more reliable because the prediction depends on actual readiness rather than assumptions.

Compliance formula

Compliance % = Passed mandatory requirements / Total mandatory requirements x 100. Gaps Found = mandatory requirements marked Gap. If evidence is still missing, recommendation remains Needs Evidence Review.

8. Win Probability Model

The current demo uses a hybrid client-side model. It is not a heavy trained Random Forest or XGBoost model. It is an explainable bid-intelligence model built around the 120 historical bid records and the current RFP compliance status.

Input Feature	Purpose
Sector win rate	Shows how often past bids in the same/related sector were won.
Compliance %	Measures how many extracted requirements are satisfied.
Gaps found	Counts open compliance issues and mandatory failures.
Proposal quality score	Estimated from compliance and capability coverage.
Budget alignment	Compares known budget against historical sector budget ranges; shows N/A when budget is missing.
Response/prep time	Estimates how much time remains before deadline.
Document pages	Proxy for opportunity complexity.
Similar past bids	k-NN style nearest historical bids based on weighted feature distance.

The engine calculates sector win rate, k-NN probability from similar bids, and a logistic-style baseline based on feature importance/correlation with historical wins. The final score combines these signals and applies penalties for mandatory gaps. If evidence is pending, the displayed probability becomes Pending Evidence instead of an artificial percentage.

Scoring concept used in the demo

k-NN probability and logistic-style probability are blended into an ensemble. The final score combines the ensemble with sector win rate, then subtracts mandatory-gap penalties and bounds the output. If compliance evidence is still missing, the percentage is hidden and the recommendation becomes Needs Evidence Review.

1. Normalize current bid features and historical bid features.

2. Calculate feature importance using correlation with historical Win/Loss outcomes.
3. Find nearest historical bids using weighted distance and sector affinity.
4. Calculate k-NN win probability from nearby historical outcomes.
5. Calculate logistic-style baseline from signed feature deviations.
6. Combine ensemble score with sector win rate and mandatory gap penalties.
7. Return probability, risk, recommendation, similar bids, and improvement suggestions.

Why not only Random Forest?
The dataset has only 120 rows, so a heavy model could overfit and be difficult to explain in a hackathon demo. The current hybrid model is transparent, lightweight, runs in the browser, and shows why the recommendation changed. In production, the same features can be moved to a backend and trained using Random Forest, XGBoost, or calibrated logistic regression.

9. LLM Draft Response Engine

The application uses Claude Sonnet 4.5 for narrative proposal drafting. The LLM is used after parsing and compliance extraction, not as the sole decision engine. This separation improves reliability: deterministic code handles mandatory requirements, while Claude handles writing quality.

LLM Feature	Implementation
API key	Stored in .env.local as VITE_CLAUDE_API_KEY for demo.
Model	VITE_CLAUDE_MODEL=claude-sonnet-4-5.
Prompt context	RFP metadata, sector, question, requirements, evidence state, and no-fabrication rules.
No fabrication rule	Claude must not invent names, certifications, references, percentages, or project details. Missing details become placeholders.
Fallback	If API fails, offline templates generate domain-specific draft text.

The no-fabrication rule is important because proposal drafts must be safe. For example, if team names or client references are not uploaded, the draft uses placeholders such as [Project Manager Name], [Cisco Certification], [Client Reference], or [Evidence Attachment] instead of inventing content.

10. Export and Workspace Management

Each uploaded RFP/RFQ creates its own workspace. Earlier demo workspaces were removed so that the dashboard no longer shows dummy win probabilities. This prevents judges or users from confusing preloaded examples with actual uploaded tender analysis.

The export pack includes opportunity overview, compliance checklist, win probability analysis, similar historical bids, recommended improvements, and draft responses. Drafts are saved using a document-specific key that includes workspace ID, title, sector, and question text, preventing draft leakage between different RFPs.

11. Why These Methods Were Used

Decision	Reasoning	Benefit
Rule-based parsing + LLM	RFPs contain tables, boilerplate, and mandatory forms that should not be hallucinated.	Stable compliance extraction with strong narrative drafting.

Evidence-aware scoring	A bidder should not get a GO recommendation before proving eligibility.	More trustworthy win probability and fewer false positives.
k-NN similar bids	Historical bids can show patterns by sector, budget, compliance, and gaps.	Explainable similar-bid table for judges and users.
Logistic-style baseline	Gives a second statistical signal without requiring a heavy model.	Balances similarity scoring and feature trend scoring.
Client-side demo	Fast hackathon setup and no backend dependency for core demo.	Easy to run locally and present quickly.
Claude for drafts	Proposal writing needs natural, professional language.	Produces complete, editable technical response sections.

12. Benefits for Bidders and Evaluators

- Reduces time spent manually reading long RFP/RFQ documents.
- Highlights mandatory evidence before the team wastes time drafting.
- Converts TOR, scope, and evaluation criteria into clear proposal questions.
- Prevents misleading GO decisions by using Pending Evidence Review when proof is missing.
- Shows similar historical bids and explainable reasons behind the score.
- Creates a downloadable bid response pack for team review and management approval.
- Improves proposal quality by generating first drafts that the bidder can edit and approve.

13. Tested Scenarios and Current Status

Document	Expected Domain	Observed / Target Behaviour
RFQ-006 Enterprise Routers	Network Infrastructure	Extracts 32 mandatory requirements, USD, deadline 18 June 2026, router-specific questions, and evidence-aware decision.
RFQ-007 Radio Campaign	Media / Communications	Extracts radio airtime requirements, PKR, station booking/monitoring/reporting evidence, and radio-specific drafts.
RFP-166 RAAST Awareness Campaign	Media / Communications	Should extract video production, radio production, creative capability, paid social, client references, and team profile requirements.
Sample RFP-031 IT Services	IT Services	Extracts ISO 27001, CMMI L3, cloud projects, data residency, ERP APIs, service desk, training, and IPR requirements.

The latest version removes preloaded sample workspaces, supports document-specific draft storage, and keeps win probability pending until evidence is completed. The system is demo-ready for uploaded RFP/RFQ analysis, with production hardening still recommended.

14. Limitations and Future Improvements

- Current PDF extraction depends on selectable text. Scanned PDFs require OCR integration.
- The 120 historical bid records are embedded for demo; production should store bid history in a database and support dataset upload/admin management.

- Claude API should be called from a backend/FastAPI service, not directly from the browser.
- The hybrid win model is explainable but not a fully trained ML model. Future versions can use calibrated Random Forest/XGBoost with cross-validation.
- Evidence is currently marked manually. Future versions should allow document upload per requirement and automatically verify evidence using AI/NLP.
- Financial scoring can be improved by using competitor pricing, procurement method, bid security, and total cost of ownership.
- Role-based access, audit trail, and approval workflow should be added for enterprise bid teams.

15. Conclusion

BidEngine demonstrates a complete AI-assisted bid intelligence workflow. It does not simply summarize documents; it extracts mandatory requirements, guides the bidder to complete evidence, creates proposal response drafts, and provides an explainable win probability based on compliance readiness and historical bid outcomes.

The strongest design choice is separating deterministic compliance logic from LLM writing. This makes the system more trustworthy: the LLM helps produce professional drafts, while the win/no-bid decision is driven by structured evidence, gaps, and historical patterns.

Final positioning

BidEngine is best presented as an AI-powered Bid/No-Bid decision-support and proposal drafting platform that combines RFP parsing, compliance evidence tracking, historical bid analytics, and LLM-assisted response generation.

Appendix A - Suggested Demo Script

- Upload an RFP/RFQ document.
- Show that the engine extracts client, title, sector, deadline, budget, currency, and selection method.
- Open the compliance checklist and explain that every mandatory item needs evidence.
- Show Pending Evidence Review before evidence is completed.
- Mark evidence as provided or mark gaps to demonstrate recalculation.
- Generate Claude draft responses for key proposal questions.
- Export the full bid response pack.
- Explain that preloaded dummy workspaces were removed; only uploaded documents create analysis.